Assessment of Knowledge Graph Construction Techniques of Patient Operation Reports

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```python
# Imports:
import random
from threading import Thread
from time import sleep, time
from os import startfile, system, path
from global_hotkeys import register_hotkeys, start_checking_hotkeys
from bot import click_if_exists, search_and_click, found, find, bclick
from input_boxes import message, buttons, double_input, loading_bar, loading_bar_done
from pyautogui import hotkey, moveTo
from pyperclip import copy
from subprocess import call
from win32gui import MoveWindow, FindWindow
from tendo.singleton import SingleInstance

# Startup Process:
'''Checks to make sure there is only one instance of the program. Then, resizes the window, and moves it to the botto
only_one = SingleInstance()
try:
    hwnd = FindWindow(None, "MagicART-Hotkeys.py - Shortcut")
    MoveWindow(hwnd, -7, 748, 407, 300, True)
except Exception as e:
    pass

# Global Variables:
awake = True
keyboard_command = True
admin = False

# Login:
'''Asks the user to login. If the user does not have an account, it asks the user if they would like to create one.
If the user is an admin, it sets the admin variable to True. Runs in response to the check_user() function.'''


def login():
```

Research Focus

Extract valuable information from Operation Reports automatically and accurately

Motivation

- Lack of interoperability between institutions
  - Create a structured representation of information so that it can be understood between institutions
- Unstructured data requires clinical coders
  - Inaccurate ICD codes hinder funding & care
- Create a structured representation of information so that diagnoses & patient journeys can be consistently captured (support audits, research, funding)
- Increase trust and confidence in LLM responses within the medical context with knowledge graphs
  - GraphRAG
  - MedGraphRAG

Data

- Patient Records from Perth Children's Hospital (PCH)
  - Admission Forms
  - Progress Notes
  - Discharge Summaries
  - Operation Reports

- Over 6000 Operation Reports with a valid entry specifying the principal diagnosis represented by an ICD code
  - This classification will base the performance

Operation Report

(Patient and Surgeon Information)

Operation and Principal Diagnosis

(Diagnosis, Findings, Procedure)

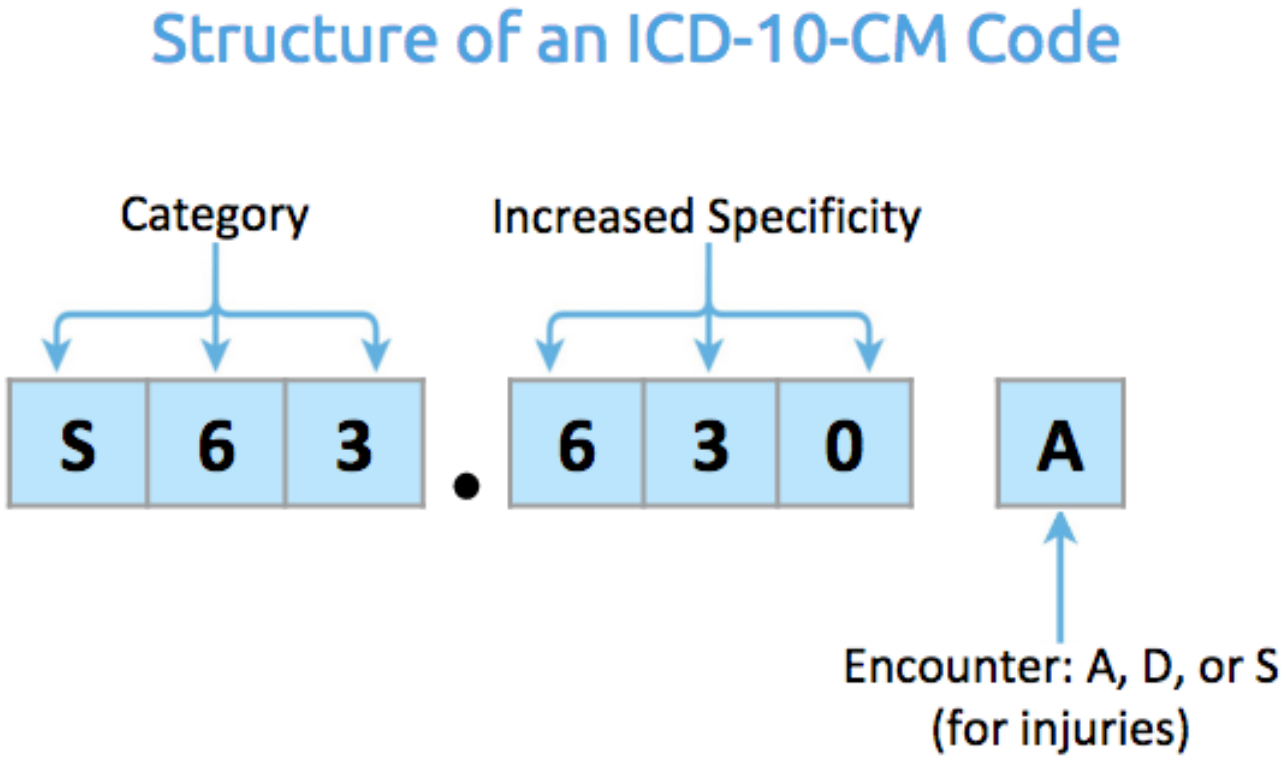
Authorisation

(Signature / Date)

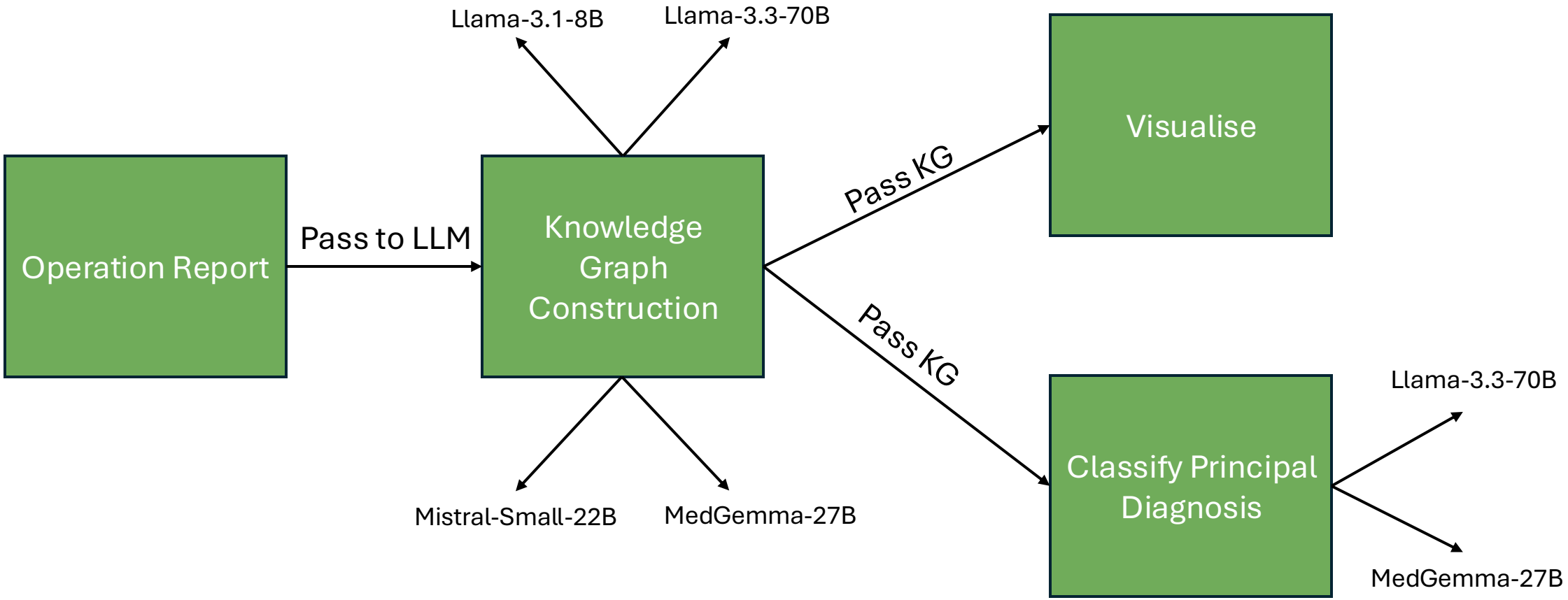
Structure of an Operation Report

ICD-10-CM

- International Classification of Diseases (ICD), Tenth Revision, Clinical Modification (ICD-10-CM)
- Performance is measured based on the classification of the Principal Diagnosis stored as an ICD-10-CM code



Experimental Design



Prompting Techniques for Knowledge Graph Construction

Zero-Shot	Few-Shot	Few-Shot + Ontology
KGGen (Knowledge Graph Generation)	EDC (Extract Define Canonicalize)	Text2KGBench (Text to Knowledge Graph Benchmark)

KGGen: Zero-Shot Prompt

Prompt for extracting entities: Extract key entities from the given text. Extracted entities are nouns, verbs, or adjectives, particularly regarding sentiment. This is for an extraction task, please be thorough and accurate to the reference text.

Prompt for extracting relations: Extract subject-predicate-object triples from the assistant message. A predicate (1-3 words) defines the relationship between the subject and object. Relationship may be fact or sentiment based on assistant's message. Subject and object are entities. Entities provided are from the assistant message and prior conversation history, though you may not need all of them. This is for an extraction task, please be thorough, accurate, and faithful to the reference text.

→ Named Entity Recognition

→ Relation Extraction

EDC: Few-Shot Prompt

Given a piece of text, extract relational triplets in the form of [Subject, Relation, Object] from it. Here are some examples:

Example 1:
Text: The 17068.8 millimeter long ALCO RS-3 has a diesel-electric transmission.
Triplets: [['ALCO RS-3', 'powerType', 'Diesel-electric transmission'], ['ALCO RS-3', 'length', '17068.8 (millimetres)']] ...

Now please extract triplets from the following text: Alan Shepard was born on Nov 18, 1923 and selected by NASA in 1959. He was a member of the Apollo 14 crew.

Example parsed with the initial prompt (Few-Shot Prompting)

Text2KGBench: Few-Shot Prompt

Given the following ontology, examples and sentences, please extract the triples from the sentence according to the relations in the ontology. In the output, only include the triples in the given output format.

CONTEXT:

Ontology Concepts: *human, city, country, film, film genre, film production company, film award, award, written work, film character, film organization*

Ontology Relations: cast_member(film,human), director (film,human), screenwriter (film,human), producer(film,human), genre(film,genre), based_on(film,written work), award_received (film,award), production_company(film,film production company), country_of_origin(film,country), publication_date (film,date), characters(film,film character), narrative_location(film,city), filming_location(film,city), main_subject(film,thing), nominated_for(film,award), cost(film,number)

Example Sentence*: The Lion King is a animated musical drama film about a lion cub who is to succeed his father and it was directed by Roger Allers and Rob Minkoff (in their feature directorial debuts), produced by Don Hahn.*

Example Output: director(Lion King, Roger Allers)
director(Lion King, Rob Minkoff)
producer(Lion King, Don Hahn)

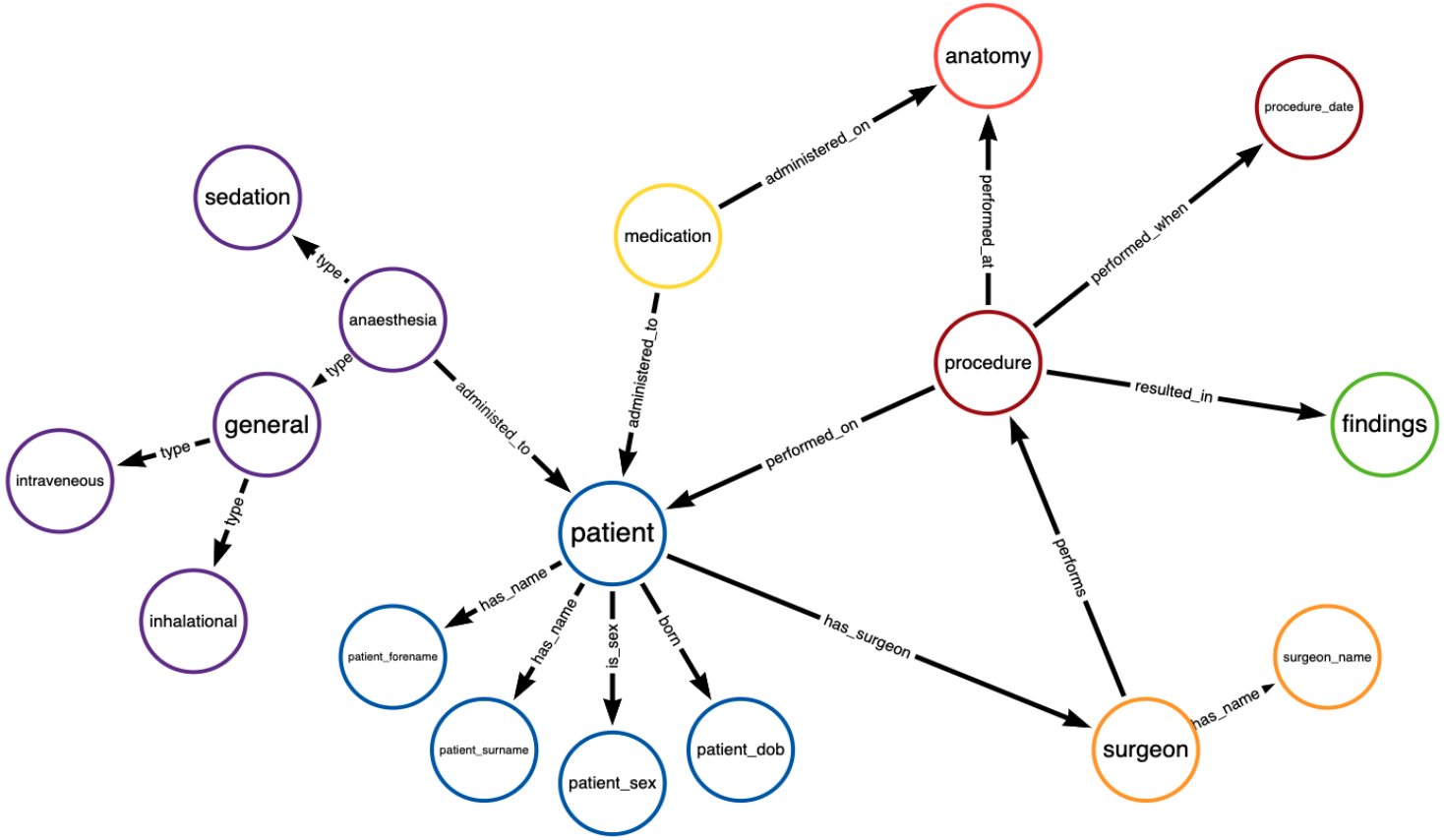
Test Sentence: Birds Anonymous is a 1957 Warner Bros. Merrie Melodies animated short, directed by Friz Freleng and written by Warren Foster.

Test Output:

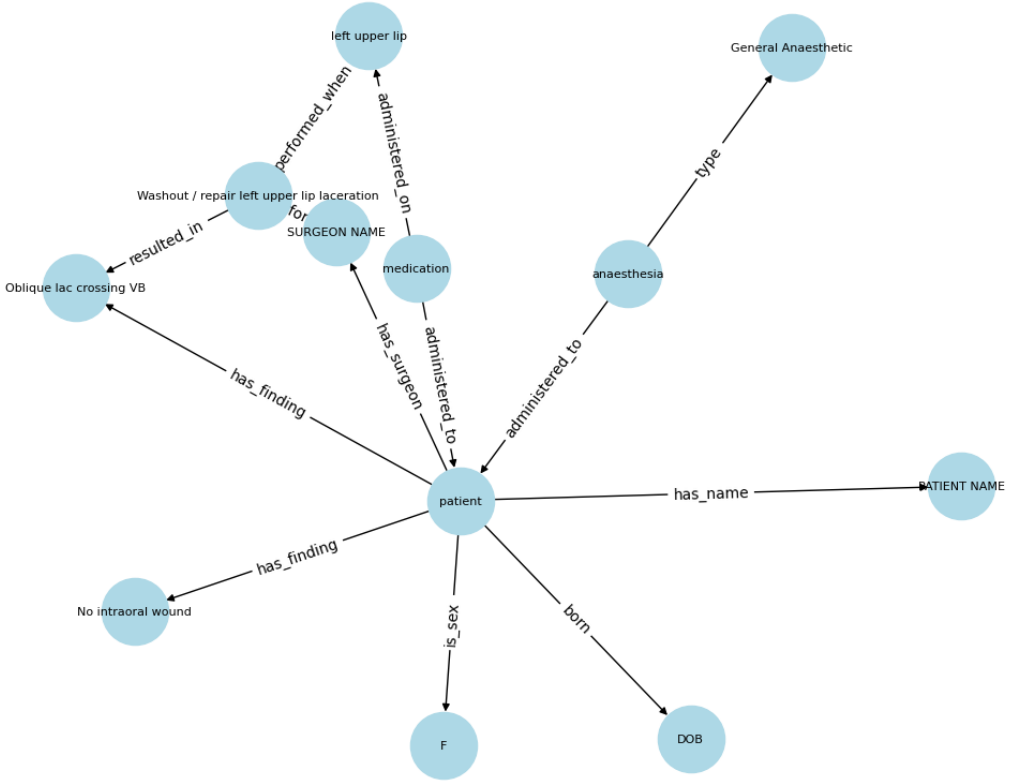
Pre-defined Ontology structure

Example parsed within the prompt (Few-Shot Prompting)

Text2KGBench: Ontology



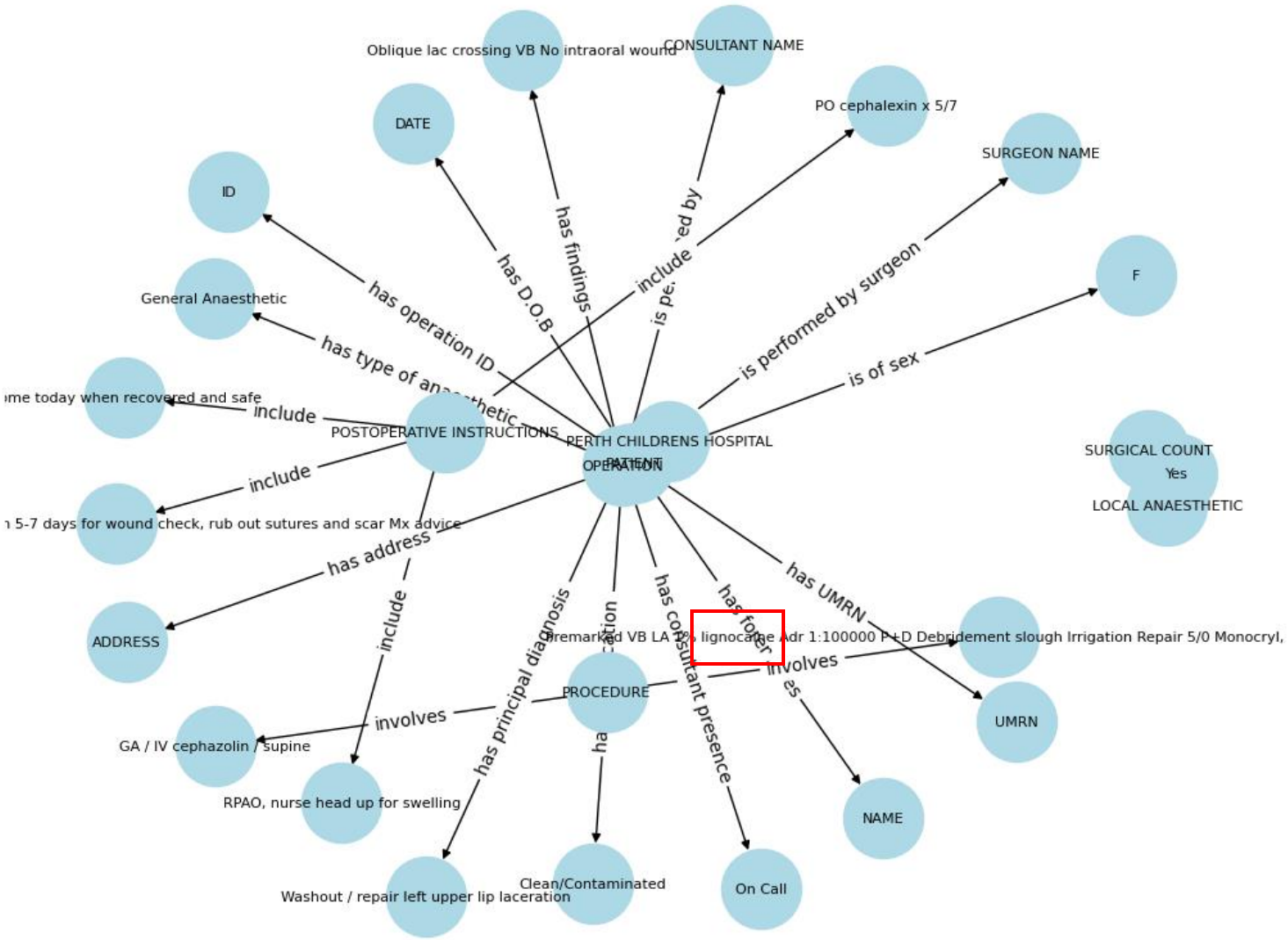
Constructed Knowledge Graphs



Text2KGBench

EDC

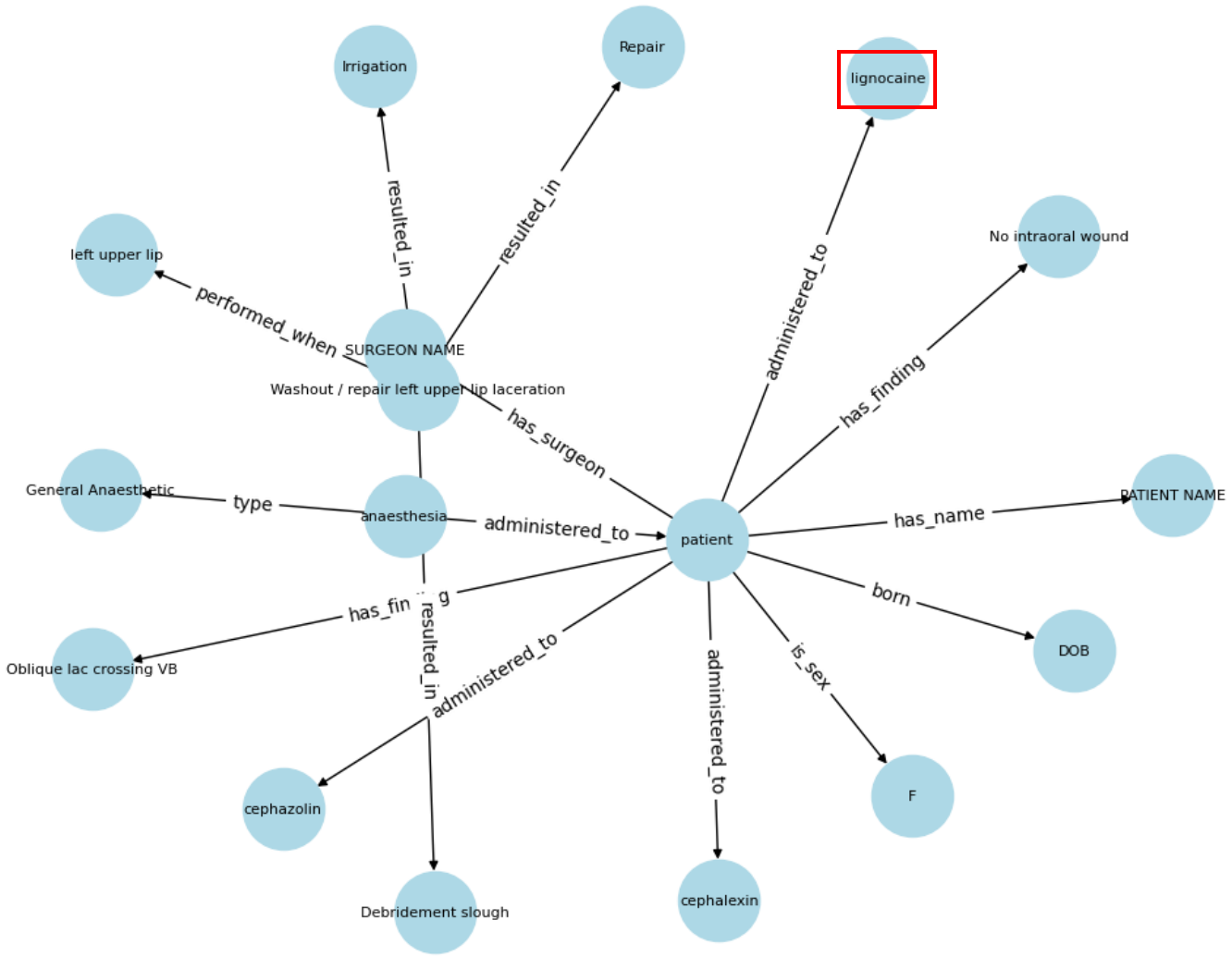
Constructed Knowledge Graphs



KGGen

Constructed Knowledge Graphs

- MedGemma as LLM for KGC
- Text2KGBench for the prompt design



Using the graph generated by a medically-tuned LLM we get additional information seen in KGGen such as the anaesthetics and medication.

Target Metrics to Assess Performance

Accuracy of Classification:

- Initial Category Match: "**S**"
- Category : "**S52**"
- Complete Match: "**S52.521A**"

Code	Code Description
S52	Fracture of forearm
S52.5	Fracture of lower end of radius
S52.52	Torus fracture of lower end of radius
S52.521	Torus fracture of lower end of right radius
S52.521A	Torus fracture of lower end of right radius, initial encounter

Table 1: Example of increasing specificity in ICD-10 code structure

Results: Baseline (Plaintext Classification)

	Llama-3-70B	MedGemma-27B
Exact Match	0.11	0.024
Category Match	0.46	0.219
Initial Category Match	0.656	0.445

Results: Exact Match

Baseline

Llama-3-70B	MedGemma-27B
0.11	0.024

Predictor: Llama-3-70B

	Mistral-Small-22B	Llama-3-8B	Llama-3-70B	MedGemma-27B
KGGen	**0.092**	**0.088**	0.094	**0.1**
EDC	0.087	0.06	**0.105**	0.09
Text2KGBench	0.083	0.08	0.089	0.085

Predictor: MedGemma-27B

	Mistral-Small-22B	Llama-3-8B	Llama-3-70B	MedGemma-27B
KGGen	**0.031**	**0.031**	**0.038**	**0.036**
EDC	0.028	0.02	0.033	0.033
Text2KGBench	0.017	0.025	0.017	0.027

Results: Category Match

Baseline

Llama-3-70B	MedGemma-27B
0.46	0.219

Predictor: Llama-3-70B

	Mistral-Small-22B	Llama-3-8B	Llama-3-70B	MedGemma-27B
KGGen	0.369	0.345	0.401	0.36
EDC	**0.404**	0.246	**<u>0.448</u>**	**0.424**
Text2KGBench	0.393	**0.353**	0.407	0.391

Predictor: MedGemma-27B

	Mistral-Small-22B	Llama-3-8B	Llama-3-70B	MedGemma-27B
KGGen	0.202	0.192	0.223	0.184
EDC	0.197	0.134	**0.231**	**<u>0.253</u>**
Text2KGBench	**0.214**	**0.204**	0.169	0.212

Results: Initial Category Match

Baseline

Llama-3-70B	MedGemma-27B
0.656	0.445

Predictor: Llama-3-70B

	Mistral-Small-22B	Llama-3-8B	Llama-3-70B	MedGemma-27B
KGGen	0.581	0.598	0.638	0.565
EDC	0.608	0.429	**<u>0.657</u>**	**0.628**
Text2KGBench	**0.643**	**0.623**	0.628	0.626

Predictor: MedGemma-27B

	Mistral-Small-22B	Llama-3-8B	Llama-3-70B	MedGemma-27B
KGGen	0.457	0.447	**0.492**	0.455
EDC	0.453	0.329	0.489	**<u>0.524</u>**
Text2KGBench	**0.507**	**0.497**	0.441	0.501

What does it all mean?

Utilising smaller parameter models:

- Text2kgbench performs best
    - Initial Category and Category

Larger parameter models:

- EDC performs best
    - Also outperforming the baseline with MedGemma as the ICD classifier

Smaller parameter-sized models benefit from ontology guidance

- Ontology improves the understanding of the principal diagnosis at the category level

Ontology guidance yields slightly worse results (Llama-3 classifier)

- Suggests ontology structures retain usefulness despite a minor drop

Prediction Difference

	Llama-3-70B	MedGemma-27B
Exact Match	0.11	0.024
Category Match	0.46	0.219
Initial Category Match	0.656	0.445

":** Excision of midline dermoid cyst- glabllar region #### **FINDINGS:** ~1cm rounded cyst tightly attachewd to underlying periosteum ## **PROCEDURE DETAILS:** GA, LA, IVABs Vertical midline incision Cyst excised completely with underlying attached periosteum Sent for histopathology Layered closure with 5-0 Monocryl Skin 5-0 Monocryl pull through monocryl Steristrips, Fixomull ## **LOCAL ANAESTHETIC:** Yes 3ml .2% Ropivacaine + Adrenaline **SURGICAL COUNT CORRECT:** Yes ## **POSTOPERATIVE INSTRUCTIONS:** RPAO Discharge when safe on simple analgesics Keep dressing dry 48 hrs the allowed to wet and pat dry PDC review in 10-14 days #### **DRAINS:** No **"

Llama-3-70B Prediction:

- D22.3
- Melanocytic nevi of unspecified part of face

Med-Gemma-27B Prediction:

- Q82.2
- Congenital cutaneous mastocytosis

Ground-Truth:

- D23.3
- Other benign neoplasm of skin of unspecified part of face

Why the difference in Performance?

	Llama-3-70B	MedGemma-27B
Exact Match	0.11	0.024
Category Match	0.46	0.219
Initial Category Match	0.656	0.445

- Instruction-following strength
  - Llama-3-70B evaluated on IFEval (Instruction-Following Evaluation) (92.5%)

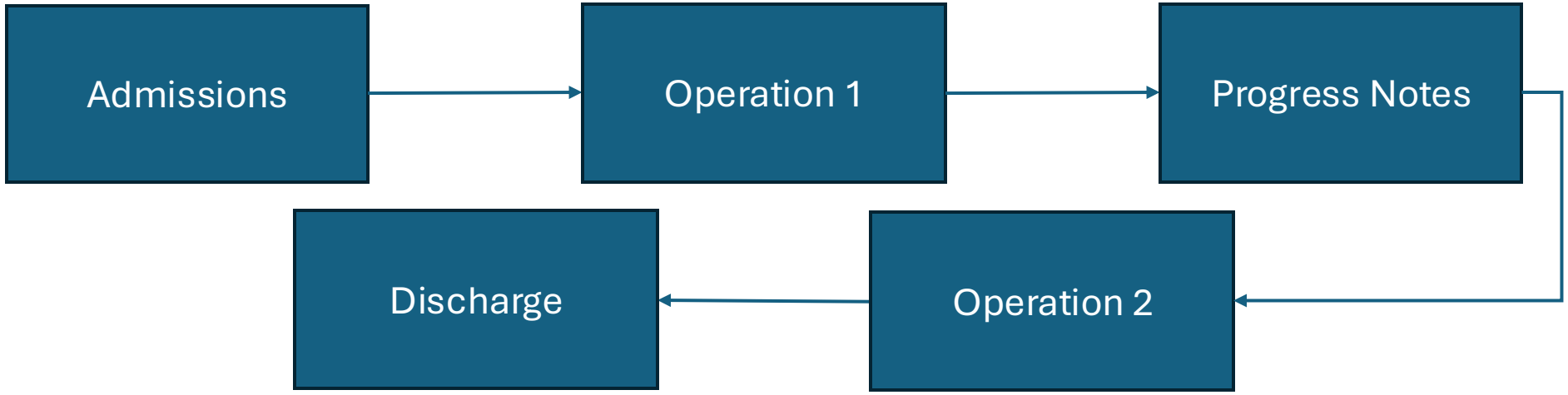
- Capacity and Scale
  - Llama-3-70B has a larger representational capacity
  - Capturing finer-grained patterns such as a Principal Diagnosis

Prompt

"You are a clinical coding assistant. Your task is to read a clinical note and return the one most appropriate ICD-10 code based on the content to describe the Principal Diagnosis. Use the official ICD-10-CM classification system."

Future Work

- Focus so far: extraction at a single time point (post-operation)
- Opportunity: capture broader patient history across visits (peri-operative)
- Next step: link information across timelines to enrich records and improve RAG accuracy



Thank You